B E N C H M A R K R E P O R T

No Gas Detected

Is Not No Gas Present When Models Trust the Meter But Not the Sludge

Behavioral Analysis of 22 LLMs on an Explosive Atmosphere Tool Selection Question

22	68.2%	0/4	6th
LLMs Tested	Reported Pass Rate	Gemma Family	R1 Parsing Issue

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1. Executive Summary

I posed a tool selection question to 22 LLMs: a pump is vibrating in an underground digested sludge manhole, anchor bolts need tightening. Gas measurement shows no gas detected. But there is ankle-deep digested sludge at the bottom. Which wrench do you use? The correct answer is D — a brass wrench, because brass is non-sparking, and digested sludge contains trapped methane that can be released by any disturbance.

The reported pass rate is 68.2% (15/22). The adjusted rate, accounting for DeepSeek R1's parsing issue, is 72.7% (16/22). This is the eighth benchmark in the WWTP series, and it introduces a trap that exploits a specific cognitive shortcut: trusting an instrument reading over environmental reasoning. Seven models chose tools that create sparks in a potentially explosive atmosphere — a decision that, in real confined space incidents, has caused fatalities.

After eight benchmarks, no model has passed every question in the series. The cumulative funnel: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0. The funnel has been at zero for four consecutive benchmarks.

F1 — The "False Safe" Trap Worked: The phrase "no gas detected at the moment" appears to have anchored 7 models on the assumption that the environment was safe. None of the 7 models that failed mentioned spark risk, methane, or explosion potential in their reasoning. They saw "no gas" and stopped thinking about atmospheric hazards entirely.

F2 — Two Wrong Answers, Two Different Blind Spots: Three models chose A (stainless steel) because they focused on corrosion resistance — "sludge is corrosive, stainless is best." Three chose C (carbon steel) because they focused on practicality — "standard tool, strong, readily available." Both reasoning paths are internally logical but miss the critical dimension: explosion risk.

F3 — Gemma Family: 0/4 — Complete Collapse: All four Gemma models failed. The three smaller models (1B, 4B, 12B) chose C; the largest (27B) chose A. Not a single Gemma model mentioned sparks, methane, or explosive atmosphere. This is the first time a family has scored 0% since DeepSeek on the Confined Space benchmark (#6), and it is Gemma's worst performance in the series.

F4 — Gemini's Generational Split: Gemini 2.0 Flash and 2.0 Flash Lite chose A (stainless steel). Gemini 2.5 Flash, 2.5 Pro, 3 Flash Preview, and 3 Pro Preview all chose D (brass). The division is clean: older generation failed, newer generation passed. This is the most distinct generational split observed in the series.

F5 — R1's Sixth Parsing Issue: DeepSeek R1 produced a 10,363-character think block with detailed reasoning about non-sparking tools, methane risk, and ATEX regulations — then answered D. The parser read "T" from the <think> tag. This is R1's 6th parsing issue in 8 benchmarks (75%).

2. The Question

This benchmark tests a specific piece of field safety knowledge: the requirement for non-sparking tools in potentially explosive atmospheres. The scenario is grounded in real confined space operations at wastewater treatment plants, where digested sludge manholes are classified as potentially explosive environments under ATEX/IECEx regulations.

The trap is embedded in the safety conditions: “gas measurement performed: NO gas detected at the moment.” In real operations, this reading is misleading. Digested sludge contains trapped methane (CH_4) and hydrogen sulfide (H_2S) in pockets below the surface. Walking through ankle-deep sludge, or any mechanical disturbance, releases these gases immediately. A single spark from a steel wrench striking a bolt can ignite the atmosphere. Real incidents with this exact scenario have resulted in fatalities.

The question offers four options — three steel variants and one brass. With 4 options, the random correct answer probability is 25%.

Table 1: The Four Options — Three Sparking, One Safe

Option	Material	Spark Risk	Description
A	Stainless Steel	SPARKING	Excellent corrosion resistance, but creates sparks on metal impact. Chosen by corrosion-focused models.
B	Galvanized Steel	SPARKING	Zinc coating for corrosion protection, but creates sparks. Not chosen by any model.
C	Carbon Steel	SPARKING	Standard workshop tool. Strong, practical, readily available — but creates sparks. Chosen by practicality-focused models.
D	Brass	NON-SPARKING	CORRECT — Brass does not create sparks on impact. Required by ATEX regulations for potentially explosive atmospheres.

Table 1: Three of four options create sparks. The question tests whether models recognize explosion risk over corrosion resistance or tool practicality.

3. Results

Figure 1: Answer Distribution

D (Brass — Correct)	16 models (incl. R1 parsed wrong)
A (Stainless Steel)	3 models (Flash 2.0, Flash Lite, Gemma 27B)
C (Carbon Steel)	3 models (Gemma 1B, 4B, 12B)

Figure 1: The wrong answers split into two distinct clusters — corrosion-focused (A) and practicality-focused (C). No model chose B (galvanized steel).

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Haiku 4.5	D	PASS	263	3.6s	Claude

Claude Opus 4.1	D	**PASS**	191	8.2s	Claude
Claude Opus 4.5	D	**PASS**	301	8.5s	Claude
Claude Sonnet 4	D	**PASS**	148	5.3s	Claude
Claude Sonnet 4.5	D	**PASS**	260	8.9s	Claude
Gemini 2.0 Flash	A	**FAIL**	219	2.5s	Gemini
Gemini 2.0 Flash Lite	A	**FAIL**	252	3.4s	Gemini
Gemini 2.5 Flash	D	**PASS**	62	5.4s	Gemini
Gemini 2.5 Pro	D	**PASS**	280	15.5s	Gemini
Gemini 3 Flash Preview	D	**PASS**	156	6.4s	Gemini
Gemini 3 Pro Preview	D	**PASS**	143	15.5s	Gemini
Gemma 3 27B	A	**FAIL**	~300	5.1s	Gemma
Gemma 3 12B	C	**FAIL**	~300	7.4s	Gemma
Gemma 3 4B	C	**FAIL**	~280	3.6s	Gemma
Gemma 3 1B	C	**FAIL**	1	1.2s	Gemma
DeepSeek R1 (CoT)	D†	**FAIL**	2,239	15.3s	DeepSeek
DeepSeek V3.1	D	**PASS**	128	2.2s	DeepSeek
DeepSeek V3.2	D	**PASS**	122	8.8s	DeepSeek
Qwen3 235B A22B Instruct	D	**PASS**	129	2.1s	Qwen
Qwen3 Coder 480B	D	**PASS**	125	2.3s	Qwen
Qwen3 Next 80B Instruct	D	**PASS**	145	2.1s	Qwen
Qwen3 Next 80B Thinking	D	**PASS**	1,296	10.8s	Qwen

Table 2: †R1 answered D inside a <think> block. Parser read "T" as first character. This is R1's 6th parsing issue in 8 benchmarks.

A note on the parsing: the Kaggle Benchmarks platform evaluates answers by extracting the first alphabetic character from each model's raw response. R1's <think> block causes the parser to read "T" instead of the intended answer. The reported pass rate (68.2%) reflects this rule; the adjusted rate (72.7%) reflects the actual answer content.

4. LLM Behavior Analysis

4.1 The "False Safe" Trap — When No Gas Detected Means No Thinking About Gas

The phrase "gas measurement performed: NO gas detected at the moment" was designed as this benchmark's primary trap. It mirrors real confined space documentation language and creates a cognitive anchor: if no gas is detected, the atmosphere is safe, and tool selection can focus on other criteria like corrosion resistance or strength.

The trap worked with precision. Of the 7 models that failed (excluding R1's parsing issue), not a single one mentioned spark risk, methane, explosive atmosphere, or the potential for gas release from disturbed sludge. Their reasoning was internally coherent — stainless steel for corrosion, carbon steel for strength — but it operated within a framework where atmospheric hazards had been entirely dismissed.

This is a qualitatively different failure mode from previous benchmarks. In the Walkway benchmark (#5), models chose FRP because they prioritized the most salient property (corrosion resistance) over the critical one (failure mode). Here, models did not even identify the critical property. They did not weigh explosion risk and dismiss it — they never considered it at all. The "no gas detected" phrase appears to have removed the entire hazard category from their reasoning.

Table 3: What Passing and Failing Models Discussed

Model (selected)	Spark	Methane	Sludge Gas	ATEX	Score
Passing Models (15)					
Claude Opus 4.5	✓	✓	✓	✗	**All 3**
Claude Sonnet 4.5	✓	✓	✓	✗	**All 3**
Gemini 2.5 Pro	✓	✓	✓	✗	**All 3**
Qwen3 235B	✓	✓	✓	✗	**All 3**
DeepSeek R1 (actual)	✓	✓	✓	✓	**All 4**
Failing Models (7)					
Gemini 2.0 Flash	✗	✗	✗	✗	**0/4**
Gemini 2.0 Flash Lite	✓	✗	✗	✗	**1/4**
Gemma 3 27B	✗	✗	✗	✗	**0/4**
Gemma 3 12B	✗	✗	✗	✗	**0/4**
Gemma 3 4B	✗	✗	✗	✗	**0/4**
Gemma 3 1B	✗	✗	✗	✗	**0/4**

Table 3: Representative models shown; full results in Table 2. The divide is absolute. Passing models identified the hazard; failing models did not. Only R1 mentioned ATEX regulations specifically.

4.2 Two Paths to Failure — Corrosion vs. Practicality

The 6 wrong answers (excluding R1) split into two distinct reasoning clusters, each internally coherent but missing the same critical dimension:

The corrosion cluster (A — stainless steel) consisted of Gemini 2.0 Flash, Gemini 2.0 Flash Lite, and Gemma 3 27B. Their reasoning focused on the digestive sludge environment: "Digested sludge is highly corrosive due to the presence of sulfides." "Stainless steel offers the best corrosion resistance." This logic is not wrong — sludge is indeed corrosive — but it addresses

the wrong problem. Corrosion affects tool longevity over months; a spark affects survival in milliseconds.

The practicality cluster (C — carbon steel) consisted of Gemma 3 1B, 3 4B, and 3 12B. Their reasoning focused on mechanical requirements: “Carbon steel provides the necessary strength and durability.” “Standard and most practical choice.” This is also not wrong in isolation — carbon steel wrenches are indeed the standard workshop tool — but it treats the manhole as an ordinary workspace rather than a potentially explosive atmosphere.

What unites both clusters is not what they said, but what they did not say. Neither cluster mentioned sparks, methane, explosive risk, or the possibility that disturbing sludge could release trapped gases. The “no gas detected” phrase appears to have functioned as a permission to ignore atmospheric hazards entirely.

Table 4: The Two Failure Clusters

Cluster	Answer	Reasoning Focus	What Was Missed
Corrosion	A (Stainless Steel)	Sludge is corrosive → best corrosion resistance	Spark = explosion in trapped methane
Practicality	C (Carbon Steel)	Standard tool → strong, available, practical	Spark = explosion in trapped methane

Table 4: Two different wrong answers driven by two different valid-but-incomplete reasoning frameworks. Both missed the same lethal dimension.

4.3 R1’s Parsing Pattern — Sixth Occurrence in Eight Benchmarks

DeepSeek R1 produced the most thorough reasoning of any model on this benchmark: a 10,363-character think block that discussed methane risk, non-sparking tool requirements, ATEX regulations, and the specific danger of disturbing sludge. It was the only model to mention ATEX by name. Its answer was D — correct.

The parser read “T” from the <think> tag. This is R1’s 6th parsing issue in 8 benchmarks, raising the occurrence rate to 75%. In 3 of those cases, the correct answer was lost to formatting.

Table 5: R1 Parsing History Across the WWTP Series

Benchmark	Raw Char	Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> block → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> block + wrong answer
#4 Biogas	—	—	—	(data pending)
#5 Walkway Material	*	B	No (FAIL)	**B** markdown
#6 Confined Space	*	O	Yes (FAIL*)	**O** markdown → parsed wrong

#7 Sample Preservation	<	D	Yes (FAIL*)	<think> block → T parsed
#8 Tool Selection	<	D	Yes (FAIL*)	<think> block → T parsed

Table 5: R1 has encountered parsing issues in 6 of 8 benchmarks. In 4 cases, the correct answer was lost to formatting. The <think> block pattern has returned after benchmarks 5–6 used bold markdown.

An ironic dimension emerges: on this benchmark, R1 produced the deepest and most comprehensive safety reasoning of all 22 models — and was the only model scored as failed despite getting the answer right. The model with the best understanding received the worst score.

4.4 Family Behavioral Patterns

Figure 2: Family Pass Rates

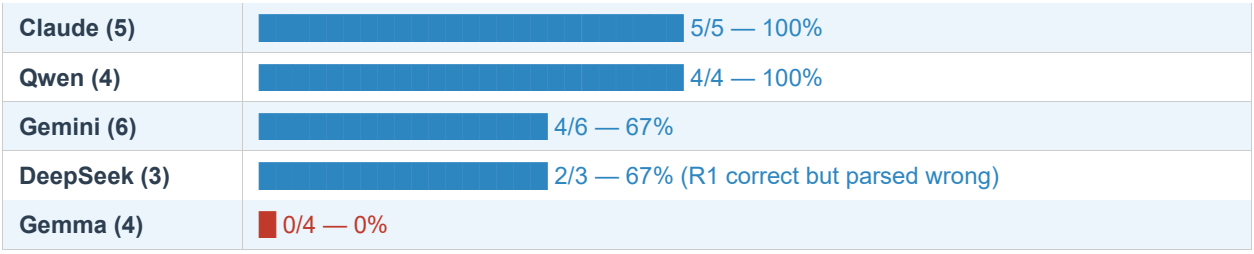

Figure 2: Claude and Qwen maintain their streak at 100%. Gemma collapses to 0% — the worst family performance on any benchmark since DeepSeek's 0/3 on Confined Space (#6).

The Gemini generational split deserves attention. The two 2.0-generation models (Flash and Flash Lite) chose stainless steel for corrosion resistance. The four newer models (2.5 Flash, 2.5 Pro, 3 Flash Preview, 3 Pro Preview) all identified the non-sparking requirement. This suggests that the knowledge of explosive atmosphere tool requirements may have been added or strengthened in training data between the 2.0 and 2.5 model generations.

Gemma's 0/4 is notable in context. In the previous benchmark (Sample Preservation, #7), all four Gemma models answered correctly — including Gemma 1B, which had given the most dangerous answer in the series on Confined Space (#6). The oscillation between perfect scores and complete failure reinforces the observation that smaller model families are unpredictable across question types rather than consistently weak.

4.5 Trap Taxonomy Update — Eight Benchmarks, Eight Failure Modes

This benchmark's "False Safe" trap shares a structural similarity with the Confined Space benchmark (#6): both exploit a gap between surface-level safety information and deeper operational reality. In #6, the trap was protocol completeness (7/8 elements feels sufficient). Here, the trap is instrument reading reliance ("no gas" feels safe). Both test whether models can reason beyond what is explicitly stated as safe.

Table 6: Updated Trap Taxonomy — Eight Benchmarks

Benchmark	Trap Name	The Shortcut	Pass %	What Was Missed
#1 Material Selection	—	—	100%	—
#2 Pump RCA	Electrical Trap	High current = winding damage	77%	Mechanical overload
#3 Dewatering RCA	Local Trap	Vibration = bearings	68%	Upstream grit system
#4 Biogas	Aerobic Trap	SOB = aerobic → aeration	14%	Digester sulfur cycling
#5 Walkway	Corrosion Trap	H_2S = best corrosion resist.	36%	Brittle failure at height
#6 Confined Space	"Almost Safe"	Strong protocol = sufficient	68%	Engineer's presence
#7 Sample Preservation	Acid Trap	Acid preserv. = always correct	95.5%	Unknown → general preserv.
#8 Tool Selection	"False Safe"	No gas detected = safe	68.2%	Trapped gas in sludge

Table 6: Eight benchmarks, eight different cognitive challenges. The "False Safe" trap is the first in the series to exploit instrument reading reliance rather than knowledge gaps or reasoning shortcuts.

5. Cross-Benchmark Comparison — Eight Questions Deep

Figure 3: Pass Rate Across 8 WWTP Benchmarks

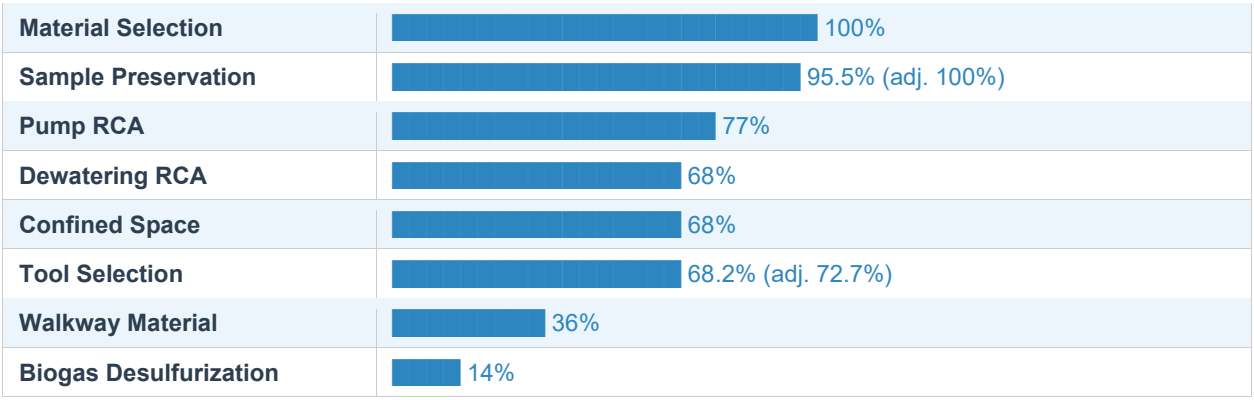

Figure 3: Eight benchmarks spanning 14% to 100%. Tool Selection sits at the same pass rate as Dewatering and Confined Space, forming a mid-difficulty cluster.

The series now covers eight distinct cognitive dimensions. The difficulty gradient reveals three tiers: easy questions where nearly all models pass (Material Selection, Sample Preservation), mid-range questions where roughly two-thirds pass (Pump RCA, Dewatering, Confined Space, Tool Selection), and hard questions where most fail (Walkway, Biogas). The mid-range tier appears to be where domain-specific safety knowledge separates capable models from those relying on general reasoning.

The cumulative funnel remains at zero. After eight benchmarks, no model has passed every question in the series. The funnel progression: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0. The

funnel closed at benchmark 5 and has not reopened — each subsequent question, regardless of difficulty, has maintained at least one failure among previously clean models.

6. Conclusions

This benchmark adds an eighth dimension to the WWTP series: tool selection in potentially explosive atmospheres. The “False Safe” trap — a clean gas reading in a sludge-rich environment — divided models into those that reason about hidden hazards and those that accept surface-level safety information at face value.

The most revealing finding is not who got it right, but how the wrong answers split. The corrosion cluster (stainless steel) and the practicality cluster (carbon steel) represent two different reasoning frameworks, both valid in ordinary workshop contexts, both potentially fatal in a confined space with trapped methane. Neither cluster identified the hazard. The “no gas detected” phrase did not cause models to weigh and dismiss explosion risk — it appears to have removed the entire hazard category from consideration.

Gemma’s 0/4 performance, following a 4/4 on the previous benchmark, continues a pattern of extreme variability in smaller model families. The oscillation between perfect and zero suggests that these models’ safety-critical knowledge is inconsistently activated rather than absent. For practical applications, inconsistency may be a greater risk than consistent failure — a model that is sometimes right and sometimes dangerously wrong is harder to trust than one that is reliably limited.

R1’s parsing issue now appears in 75% of all WWTP benchmarks. On this particular question, the irony is sharpest: R1 produced the most thorough safety reasoning of any model, including the only explicit mention of ATEX regulations, yet received the worst score. The gap between reasoning quality and evaluation outcome has never been wider.

A broader observation: of the 15 models that answered correctly, most identified the non-sparking requirement through general safety reasoning. Only R1 referenced the specific regulatory framework (ATEX/IECEx) that governs tool selection in explosive atmospheres. This suggests that while many models have absorbed the practical principle — brass tools in gas-risk environments — the regulatory knowledge underpinning that principle appears to be sparsely represented. The gap is not just in hazard recognition but in regulation-level domain knowledge.

The series continues at eight benchmarks with zero models maintaining a perfect record. The funnel at zero for four consecutive benchmarks suggests that the series has found a difficulty level that reliably differentiates — not between good and bad models, but between models with and without specific domain knowledge on each question.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute.

Single question: This report analyzes one question with 4 options. The random correct answer probability is 25%.

Safety context: The correct answer (D) is based on ATEX/IECEx regulations for non-sparking tools in potentially explosive atmospheres. The scenario reflects real operational conditions in digested sludge manholes where trapped methane poses continuous ignition risk regardless of momentary gas readings.

Parsing: DeepSeek R1's response begins with <think>. The adjusted pass rate (72.7%) accounts for this. R1's parsing behavior is tracked as a systematic pattern across the series, now at 6 of 8 benchmarks (75%).

Token counts: Some Gemma models reported 0 output tokens in API metrics despite producing visible responses. Token counts for these models are estimated from response character length and marked with ~.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.

Series consistency: All eight benchmarks in the WWTP series use the same platform, the same first-character parser, and the same default temperature settings. This ensures internal consistency across the series and allows cross-benchmark comparisons to reflect question difficulty rather than methodological variation.